US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250284430
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Lee; Seungcheol

AUTOMOTIVE STORAGE DEVICE INCLUDING AUTOMOTIVE STORAGE CONTROLLER, AND AUTOMOTIVE ELECTRONIC SYSTEM INCLUDING THE SAME

Abstract

An automotive storage device includes: a nonvolatile memory device configured to store data; and a storage controller configured to receive a first request from a first host, control the nonvolatile memory device so that the nonvolatile memory device performs an operation corresponding to the first request by using a first function including a first physical function, receive a second request from a second host having a safety integrity level that is different from that of the first host, and control the nonvolatile memory device so that the nonvolatile memory device performs an operation corresponding to the second request by using a second function including a second physical function.

Inventors: **Lee; Seungcheol (Suwon-si, KR)**

Applicant: **Samsung Electronics Co., Ltd.** (Suwon-si, KR)

Family ID: **96949238**

Assignee: **Samsung Electronics Co., Ltd. (Suwon-si, KR)**

Appl. No.: **18/894402**

Filed: **September 24, 2024**

Foreign Application Priority Data

KR	10-2024-0031438	Mar. 05, 2024

Publication Classification

Int. Cl.: **G06F3/06** (20060101)

U.S. Cl.:

CPC **G06F3/0655** (20130101); **G06F3/0604** (20130101); **G06F3/0679** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to and the benefit under 35 U.S.C. § 119(a)-(d) of Korean Patent Application No. 10-2024-0031438 filed on Mar. 5, 2024, in the Korean Intellectual Property Office, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to an automotive storage device including an automotive storage controller, and an automotive electronic system including the same.

BACKGROUND

[0003] A vehicle may include an automotive storage device that stores data generated while the vehicle (e.g., automobile) is being driven and host devices that control the automotive storage device. Host devices may have different safety integrity levels (SIL) depending on the safety required for operations performed in the vehicle. The automotive storage device may need to process requests received from the host device with a high safety integrity level with no more errors than requests received from a host device with a low safety integrity level.

SUMMARY

[0004] The present disclosure attempts to provide a storage device including a storage controller for preventing errors of functions used to requests received from an outside, and an electronic system including the same.

[0005] An embodiment of the present disclosure provides an automotive storage device including: a nonvolatile memory device configured to store data; and a storage controller configured to receive a first request from a first host, control the nonvolatile memory device so that the nonvolatile memory device performs an operation corresponding to the first request by using a first function including a first physical function, receive a second request from a second host having a safety integrity level that is different from that of the first host, and control the nonvolatile memory device so that the nonvolatile memory device performs an operation corresponding to the second request by using a second function including a second physical function.

[0006] Another embodiment of the present disclosure provides an automotive storage controller including: first registers configured to store a first function including a first physical function; second registers configured to store a second function including a second physical function; and a processor configured to receive a first request corresponding to a first safety integrity level from outside the automotive storage controller, generate a command for performing an operation corresponding to the first request based on the first function, receive a second request corresponding to a second safety integrity level from outside the automotive storage controller, generate a command for performing an operation corresponding to the second request based on the second function, and detect generation of errors of the second function while accessing the second registers.

[0007] Another embodiment of the present disclosure provides an automotive electronic system including: a first host processor with a first safety integrity level; a second host processor with a second safety integrity level that is higher than the first safety integrity level; and an automotive storage device configured to perform an operation corresponding to a first request of the first host processor based on a first function including a first physical function in response to the first request, and perform an operation corresponding to a second request of the second host processor based on a second function including a second physical function in response to the second request.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** shows a vehicle including an electronic system according to an embodiment.

[0009] FIG. **2** shows an electronic system including a center controller according to an embodiment.

[0010] FIG. **3** shows a register group for storing functions according to an embodiment.

[0011] FIG. **4** shows a storage device for performing an operation that corresponds to a first request of a first host device according to an embodiment.

[0012] FIG. **5** shows a storage device for performing an operation that corresponds to a second request of a second host device according to an embodiment.

[0013] FIG. **6** shows a storage device for performing an operation that corresponds to a third request of a third host device according to an embodiment.

[0014] FIG. **7** shows a register including a bit flip correcting circuit according to an embodiment.

[0015] FIG. **8** shows a register group for storing a mirror function according to an embodiment.

[0016] FIG. **9** shows a storage device for detecting errors of a function by use of a mirror function according to an embodiment.

[0017] FIG. **10** shows a flowchart of a storage device for performing an operation that corresponds to a request of a host device by using functions including physical functions according to an embodiment.

[0018] FIG. **11** shows a configuration of a storage controller according to an embodiment.

[0019] FIG. **12** shows a configuration of a nonvolatile memory device according to an embodiment.

DETAILED DESCRIPTION

[0020] The present disclosure will be described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the disclosure are shown. As those skilled in the art would realize, the described embodiments may be modified in various different ways, all without departing from the spirit or scope of the present disclosure.

[0021] Parts that are irrelevant to the description will be omitted to clearly describe the present disclosure, and the same elements will be designated by the same reference numerals throughout the specification.

[0022] Unless explicitly described to the contrary, the word “comprise”, and variations such as “comprises” or “comprising”, will be understood to imply the inclusion of stated elements but not the exclusion of any other elements.

[0023] FIG. **1** shows a vehicle including an electronic system according to an embodiment.

[0024] Referring to FIG. **1**, the vehicle **10** may include an electronic system **50**. The electronic system **50** may process data generated in the vehicle **10**. The electronic system **50** may include a first zonal controller **100**, a second zonal controller **200**, a third zonal controller **300**, a fourth zonal controller **400**, a center controller **500**, and micro control units (MCUs) **101**, **102**, **201**, **202**, **301**, **302**, **401**, and **402**.

[0025] The MCUs **101**, **102**, **201**, **202**, **301**, **302**, **401**, and **402** may control operations performed by the vehicle based on information obtained from inside and/or outside of the vehicle. In an embodiment, the MCUs **101**, **102**, **201**, **202**, **301**, **302**, **401**, and **402** may control rotation rates of a motor based on information received from an acceleration pedal and a brake pedal. In an embodiment, the MCUs **101**, **102**, **201**, **202**, **301**, **302**, **401**, and **402** may control heating wires of sheets in the vehicle based on temperature information received from a temperature sensor in the vehicle.

[0026] The first zonal controller **100**, the second zonal controller **200**, the third zonal controller **300**, and the fourth zonal controller **400** may control the MCUs **101**, **102**, **201**, **202**, **301**, **302**, **401**,

and **402**.
[0027] In an embodiment, the first zonal controller **100** may receive data from the MCUs **101** and **102** and may control operations of the MCUs **101** and **102**. In an embodiment, the second zonal controller **200** may receive data from the MCUs **201** and **202**, and may control operations of the MCUs **201** and **202**. In an embodiment, the third zonal controller **300** may receive data from the MCUs **301** and **302**, and may control operations of the MCUs **301** and **302**. In an embodiment, the fourth zonal controller **400** may receive data from the MCUs **401** and **402**, and may control operations of the MCUs **401** and **402**.
[0028] The center controller **500** may control a general operation of the electronic system **50**. In an embodiment, the center controller **500** may receive data from the first zonal controller **100**, the second zonal controller **200**, the third zonal controller **300**, and the fourth zonal controller **400**. The center controller **500** may control operations of the first zonal controller **100**, the second zonal controller **200**, the third zonal controller **300**, and the fourth zonal controller **400**.
[0029] In an embodiment, the center controller **500** may process data relating to an advanced driver assistance system (ADAS). In an embodiment, the advanced driver assistance system may include a lane departure warning (LDW), a lane keeping assist (LKA), a high beam assist (HBA), an autonomous emergency braking (AEB), a traffic sign recognition (TSR), a smart cruise control (SCC), a blind spot detection (BSD), and/or a forward collision-avoidance assist (FCA).
[0030] For example, the center controller **500** may control a speaker in the vehicle **10** to output a warning sound and notify a driver that the vehicle **10** is not in a driving lane based on lane information received from a sensor for detecting lanes.
[0031] In an embodiment, the center controller **500** may process data relating to in-vehicle infotainment (IVI). For example, the center controller **500** may control to output image information displaying movies, TV, and navigation functions through a display in the vehicle **10**.
[0032] In an embodiment, the center controller **500** may store the data received from the first zonal controller **100**, the second zonal controller **200**, the third zonal controller **300**, and the fourth zonal controller **400**. In an embodiment, the center controller **500** may store data relating to the advanced driver assistance system and the in-vehicle infotainment.
[0033] FIG. **2** shows an electronic system including a center controller according to an embodiment.
[0034] Referring to FIG. **2**, the electronic system **50** may include a center controller **500**. The center controller **500** may include a storage device **1000**, a peripheral component interconnect express (PCIe) switch **2000**, a first host device **3000**, a second host device **4000**, and a third host device **5000**.
[0035] The first host device **3000**, the second host device **4000**, and the third host device **5000** may receive data from the first to fourth zonal controllers **100**, **200**, **300**, and **400**. The first host device **3000**, the second host device **4000**, and the third host device **5000** may provide data to the storage device **1000** through the PCIe switch **2000**. In an embodiment, the first host device **3000**, the second host device **4000**, and the third host device **5000** may be a system on chip (SoC).
[0036] In an embodiment, each of the first host device **3000**, the second host device **4000**, and the third host device **5000** may have a different safety integrity level (SIL). In an embodiment, the safety integrity level may be an automotive safety integrity level (ASIL) defined by the ISO 26262.
[0037] The safety integrity level may indicate degrees of safety required on the operations performed by the first host device **3000**, the second host device **4000**, and the third host device **5000**. In an embodiment, the safety integrity level may represent the degree on which the operation performed by the host device does not malfunction. In an embodiment, the safety integrity level may represent the degree on which the operation performed by the host device does not generate errors.
[0038] In an embodiment, the first host device **3000** may have a first safety integrity level, the second host device **4000** may have a second safety integrity level that is higher than the first safety

integrity level, and the third host device **5000** may have a third safety integrity level that is higher than the first safety integrity level and is lower than the second safety integrity level.

[0039] For example, the first safety integrity level may correspond to a quality management (QM) level, the second safety integrity level may correspond to an ASIL-D, and the third safety integrity level may correspond to an ASIL-B.

[0040] In another embodiment, the first host device **3000** may have the first safety integrity level, and the second host device **4000** and the third host device **5000** may have the second safety integrity level. For example, the first host device **3000** may correspond to the QM level, and the second host device **4000** and the third host device **5000** may correspond to the ASIL-D.

[0041] In an embodiment, the first host device **3000**, the second host device **4000**, and the third host device **5000** may be required to have the safety that corresponds to the corresponding safety integrity level. In an embodiment, the first host device **3000** may perform an operation that has the safety required by the first safety integrity level. The second host device **4000** may perform an operation that has the safety required by the second safety integrity level. The third host device **5000** may perform an operation that has the safety required by the third safety integrity level.

[0042] In an embodiment, the first host device **3000** with the first safety integrity level may perform an operation requiring lower safety than the second host device **4000** with the second safety integrity level that is higher than the first safety integrity level. The second host device **4000** with the second safety integrity level may perform an operation requiring higher safety than the third host device **5000** with the third safety integrity level that is lower than the second safety integrity level. The first host device **3000** with the first safety integrity level may perform an operation requiring lower safety than the third host device **5000** with the third safety integrity level that is higher than the first safety integrity level.

[0043] For example, the first host device **3000** with the first safety integrity level that corresponds to the QM level may perform an operation for processing the data relating to the in-vehicle infotainment that is an operation requiring lower safety than the ASIL-D. The second host device **4000** with the second safety integrity level that corresponds to the ASIL-D may perform an operation for processing the data relating to control of an air bag, an anti-lock brake, and a power steering that are operations requiring higher safety than the QM level. The third host device **5000** with the third safety integrity level that corresponds to the ASIL-B may perform an operation for processing the data relating to a rear-view camera or control such as a stop that are operations requiring lower safety than the ASIL-D.

[0044] For another example, the first host device **3000** with the first safety integrity level may perform an operation for processing the data relating to the in-vehicle infotainment, and the second host device **4000** and the third host device **5000** with the second safety integrity level that is higher than the first safety integrity level may perform an operation for processing the data relating to the ADAS.

[0045] In an embodiment, the first host device **3000** may include a first host processor **3100**. The first host processor **3100** may control a general operation of the first host device **3000**. The second host device **4000** may include a second host processor **4100**. The second host processor **4100** may control a general operation of the second host device **4000**. The third host device **5000** may include a third host processor **5100**. The third host processor **5100** may control a general operation of the third host device **5000**.

[0046] In an embodiment, the first host processor **3100** may control an operation that corresponds to the first safety integrity level. The second host processor **4100** may control an operation that corresponds to the second safety integrity level. The third host processor **5100** may control an operation that corresponds to the third safety integrity level.

[0047] The PCIe switch **2000** may control data communication between the first host device **3000**, the second host device **4000**, and the third host device **5000**, and the storage device **1000**. The data generated by the first host device **3000**, the second host device **4000**, and the third host device **5000** may be provided to the storage device **1000** through the PCIe switch **2000**. The data stored in the storage device **1000** may be provided to the first host device **3000**, the second host device **4000**, and the third host device **5000** through the PCIe switch **2000**.

[0048] The storage device **1000** may include a nonvolatile memory device **1100**, a storage controller **1200**, and a volatile memory device **1300**.

[0049] The nonvolatile memory device **1100** may store data. The nonvolatile memory device **1100** may be operated in response to control by the storage controller **1200**. In an embodiment, the nonvolatile memory device **1100** may be a NAND flash memory.

[0050] The nonvolatile memory device **1100** may receive commands and addresses from the storage controller **1200**, and may perform an operation, instructed by the command, on an area selected by the address. The nonvolatile memory device **1100** may perform a program operation (or a write operation) for storing data in the area selected by the address, a read operation for reading data, or an erase operation for deleting data.

[0051] In an embodiment, the nonvolatile memory device **1100** may include storage areas. In an embodiment, the nonvolatile memory device **1100** may include a first storage area **1110**, a second storage area **1120**, and a third storage area **1130**. In an embodiment, the first storage area **1110**, the second storage area **1120**, and the third storage area **1130** may include memory blocks. In an embodiment, the first storage area **1110**, the second storage area **1120**, and the third storage area **1130** may include name spaces.

[0052] The storage controller **1200** may control a general operation of the storage device **1000**.

[0053] In an embodiment, the storage controller **1200** may control the nonvolatile memory device **1100** to perform a write operation, a read operation, or an erase operation according to requests by the first host device **3000**, the second host device **4000**, and the third host device **5000**. The storage controller **1200** may provide a write command, an address, and data to the nonvolatile memory device **1100** during the write operation. The storage controller **1200** may provide a read command and an address to the nonvolatile memory device **1100** during the read operation. The storage controller **1200** may provide an erase command and an address to the nonvolatile memory device **1100** during the erase operation.

[0054] In an embodiment, the storage controller **1200** may include a processor **1210** and a register group **1220**. The processor **1210** may control a general operation of the storage controller **1200**. The processor **1210** may control the nonvolatile memory device **1100** to perform operations that correspond to the requests received from the first host device **3000**, the second host device **4000**, and the third host device **5000** by using the functions stored in the register group **1220**.

[0055] The register group **1220** may include registers. The registers included in the register group **1220** may store a first function **1230**, a second function **1240**, and a third function **1250**. The first function **1230**, the second function **1240**, and the third function **1250** may include commands used in performing operations that correspond to the requests received from the first host device **3000**, the second host device **4000**, and the third host device **5000**.

[0056] In an embodiment, the first function **1230**, the second function **1240**, and the third function **1250** may respectively include a physical function and virtual functions. In an embodiment, the physical function and the virtual functions may support single root I/O virtualization (SR-IOV) between the first host device **3000**, the second host device **4000**, or the third host device **5000** and the storage device **1000**.

[0057] The physical function may be used to perform an operation that corresponds to the request received from the first host device **3000**, the second host device **4000**, or the third host device **5000**. The virtual functions may be used to store or read the data generated by virtual machines, operating systems, or applications performed by the first host device **3000**, the second host device **4000**, or the third host device **5000**. In an embodiment, the virtual functions may include some of the commands included in the physical function. In an embodiment, the number of the virtual functions is changeable according to the numbers of the virtual machines, operating systems, or applications executed by the first host device **3000**, the second host device **4000**, or the third host device **5000**.

[0058] In an embodiment, the processor **1210** may receive requests from the first host device **3000**, the second host device **4000**, and the third host device **5000**, and may control the nonvolatile memory device **1100** to perform operations that correspond to the requests by using the first function **1230**, the second function **1240**, and the third function **1250** including different physical functions.

[0059] In an embodiment, the processor **1210** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the first request based on the first function **1230** that includes the first physical function in response to the first request received from the first host device **3000**. The operation that corresponds to the first request may include a write operation for storing the data received from the first host device **3000** in the nonvolatile memory device **1100** or a read operation for providing the data stored in the nonvolatile memory device **1100** to the first host device **3000**.

[0060] In an embodiment, the processor **1210** may allocate the first storage area **1110** for storing the data received from the first host device **3000** based on the first function **1230**. The processor **1210** may control the nonvolatile memory device **1100** to store the data received from the first host device **3000** in the first storage area **1110** based on the first function **1230**.

[0061] In an embodiment, the processor **1210** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the second request based on the second function **1240** including the second physical function in response to the second request received from the second host device **4000**.

[0062] In an embodiment, the processor **1210** may allocate the second storage area **1120** for storing the data received from the second host device **4000** based on the second function **1240**. The processor **1210** may control the nonvolatile memory device **1100** to store the data received from the second host device **4000** in the second storage area **1120** based on the second function **1240**.

[0063] In an embodiment, the processor **1210** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the third request based on the third function **1250** including the third physical function in response to the third request received from the third host device **5000**.

[0064] In an embodiment, the processor **1210** may allocate the third storage area **1130** for storing the data received from the third host device **5000** based on the third function **1250**. The processor **1210** may control the nonvolatile memory device **1100** to store the data received from the third host device **5000** in the third storage area **1130** based on the third function **1250**.

[0065] In an embodiment, the second function **1240** and the third function **1250** may include an error detection function. The error detection function may detect generation of errors of the second function **1240** and the third function **1250**, and may be used to a recovery operation for correcting errors.

[0066] The volatile memory device **1300** may temporarily store the data provided by the first host device **3000**, the second host device **4000**, and the third host device **5000**, or may temporarily store the data read from the nonvolatile memory device **1100**. In an embodiment, the volatile memory device **1300** may be a dynamic random access memory (DRAM) or a static random access memory (SRAM). In an embodiment, the volatile memory device **1300** may be disposed inside/outside the storage controller **1200**.

[0067] FIG. **3** shows a register group for storing functions according to an embodiment.

[0068] Referring to FIG. **3**, the storage controller **1200** may include a register group **1220**. The register group **1220** may include a first register group **1221**, a second register group **1222**, and a third register group **1223**.

[0069] The first register group **1221**, the second register group **1222**, and the third register group **1223** may respectively include registers. The first register group **1221** may store the first function

1230. The first function **1230** may include a first physical function PF**1**, a first_1 virtual function VF**1_1**, a first_2 virtual function VF**1_2**, and a first_3 virtual function VF**1_3**.

[0070] In an embodiment, the first register group **1221** may include a first_1 register REGISTER **1_1**, a first_2 register REGISTER **1_2**, a first_3 register REGISTER **1_3**, and a first_4 register REGISTER **1_4**. The first_1 register REGISTER **1_1** may store the first physical function PF**1**. The first_2 register REGISTER **1_2** may store the first_1 virtual function VF**1_1**. The first_3 register REGISTER **1_3** may store the first_2 virtual function VF**1_2**. The first_4 register REGISTER **1_4** may store the first_3 virtual function VF**1_3**. FIG. **3** has shown the case in which one physical function or one virtual function is stored in one register. In an embodiment, one physical function or one virtual function may be stored in the registers.

[0071] In an embodiment, the second register group **1222** may store the second function **1240**. The second function **1240** may include a second physical function PF**2**, a second_1 virtual function VF**2_1**, a second_2 virtual function VF**2_2**, a second_3 virtual function VF**2_3**, and an error detection function **1241**. The error detection function **1241** may detect generation of errors of the second function **1240**, and may correct errors of the second function **1240**. The error detection function **1241** may allow or restrict an access to the second register group **1222** in which the second function **1240** is stored.

[0072] In an embodiment, the second register group **1222** may include a second_1 register REGISTER **2_1**, a second_2 register REGISTER **2_2**, a second_3 register REGISTER **2_3**, a second_4 register REGISTER **2_4**, and a second_5 register REGISTER **2_5**. The second_1 register REGISTER **2_1** may store the second physical function PF**2**. The second_2 register REGISTER **2_2** may store the second_1 virtual function VF**2_1**. The second_3 register REGISTER **2_3** may store the second_2 virtual function VF**2_2**. The second_4 register REGISTER **2_4** may store the second_3 virtual function VF**2_3**. The second_5 register REGISTER **2_5** may store the error detection function **1241**.

[0073] In an embodiment, the third register group **1223** may store the third function **1250**. The third function **1250** may include a third physical function PF**3**, a third_1 virtual function VF**3_1**, a third_2 virtual function VF**3_2**, a third_3 virtual function VF**3_3**, and an error detection function **1251**. The error detection function **1251** may detect generation of errors of the third function **1250**, and may correct errors of the third function **1250**. The error detection function **1251** may allow or restrict the access to the third register group **1223** in which the third function **1250** is stored.

[0074] In an embodiment, the third register group **1223** may include a third_1 register REGISTER **3_1**, a third_2 register REGISTER **3_2**, a third_3 register REGISTER **3_3**, a third_4 register REGISTER **3_4**, and a third_5 register REGISTER **3_5**. The third_1 register REGISTER **3_1** may store the third physical function PF**3**. The third_2 register REGISTER **3_2** may store the third_1 virtual function VF**3_1**. The third_3 register REGISTER **3_3** may store the third_2 virtual function VF**3_2**. The third_4 register REGISTER **3_4** may store the third_3 virtual function VF**3_3**. The third_5 register REGISTER **3_5** may store the error detection function **1251**.

[0075] FIG. **4** shows a storage device for performing an operation that corresponds to a first request of a first host device according to an embodiment.

[0076] Referring to FIG. **4**, the center controller **500** may include a first host device **3000**, a storage controller **1200**, and a nonvolatile memory device **1100**.

[0077] In an embodiment, the first host device **3000** may have the first safety integrity level. For example, the first safety integrity level may correspond to the QM level.

[0078] The first host device **3000** may include a first host processor **3100**. The storage controller **1200** may include a processor **1210** and a register group **1220**. The nonvolatile memory device **1100** may include the first storage area **1110**. The first storage area **1110** may include a first_1 storage area **1111**, a first_2 storage area **1112**, and a first_3 storage area **1113**.

[0079] In an embodiment, the first host processor **3100** may execute virtual machines, operating systems, or applications. In an embodiment, the first host processor **3100** may execute the first_1

virtual machine **3110**, the first_2 virtual machine **3120**, and the first_3 virtual machine **3130**.
[0080] In an embodiment, the first host processor **3100** may control an operation that corresponds to the first safety integrity level. For example, the operation that corresponds to the first safety integrity level may process the data relating to the in-vehicle infotainment. In an embodiment, the first host processor **3100** may transmit a first request REQ**1** for instructing to store or read the data used to the operation that corresponds to the first safety integrity level to the storage controller **1200**.
[0081] The storage controller **1200** may receive the first request REQ**1** from the first host processor **3100**. In an embodiment, the first request REQ**1** may correspond to the first safety integrity level. For example, the first request REQ**1** that corresponds to the first safety integrity level may instruct to store or read the data relating to the in-vehicle infotainment.
[0082] The storage controller **1200** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the first request REQ**1** based on the first function **1230** in response to the first request REQ**1**. In an embodiment, the operation that corresponds to the first request REQ**1** may include a write operation for storing the data received from the first host processor **3100** in the nonvolatile memory device **1100**. The operation that corresponds to the first request REQ**1** may include a read operation for providing the data read from the nonvolatile memory device **1100** to the first host processor **3100**.
[0083] In an embodiment, the processor **1210** may receive the first request REQ**1** from the first host processor **3100**. The processor **1210** may access the register group **1220** in response to the first request REQ**1**, and may obtain the first function **1230** stored in the register group **1220**. The first function **1230** may include a first physical function PF**1**, a first_1 virtual function VF**1_1**, a first_2 virtual function VF**1_2**, and a first_3 virtual function VF**1_3**.
[0084] In an embodiment, the processor **1210** may control the nonvolatile memory device **1100** to store the data received from the first host processor **3100** in the first storage area **1110** based on the first physical function PF**1** in response to the first request REQ**1**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the first storage area **1110** based on the first physical function PF**1** in response to the first request REQ**1**. The processor **1210** may provide the data read from the first storage area **1110** to the first host processor **3100** in response to the first request REQ**1**.
[0085] In an embodiment, the processor **1210** may receive the first request REQ**1** from the first_1 virtual machine **3110** executed by the first host processor **3100**. The processor **1210** may access the register group **1220** in response to the first request REQ**1** received from the first_1 virtual machine **3110**, and may obtain the first_1 virtual function VF**1_1** stored in the register group **1220**.
[0086] The processor **1210** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the first request REQ**1** based on the first_1 virtual function VF**1_1** in response to the first request REQ**1** received from the first_1 virtual machine **3110**. The processor **1210** may control the nonvolatile memory device **1100** to store the data in the first_1 storage area **1111** based on the first_1 virtual function VF**1_1**. The data generated by the first_1 virtual machine **3110** may be stored in in the first_1 storage area **1111** according to the first request REQ**1**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the first_1 storage area **1111** based on the first_1 virtual function VF**1_1**.
[0087] In an embodiment, the processor **1210** may receive the first request REQ**1** from the first_2 virtual machine **3120** executed by the first host processor **3100**. The processor **1210** may access the register group **1220** in response to the first request REQ**1** received from the first_2 virtual machine **3120**, and may obtain the first_2 virtual function VF**1_2** stored in the register group **1220**.
[0088] The processor **1210** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the first request REQ**1** based on the first_2 virtual function VF**1_2** in response to the first request REQ**1** received from the first_2 virtual machine **3120**. The processor **1210** may control the nonvolatile memory device **1100** to store the data in the first_2 storage area

1112 based on the first_2 virtual function VF**1_2**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the first_2 storage area **1112** based on the first_2 virtual function VF**1_2**.
[0089] In an embodiment, the processor **1210** may receive the first request REQ**1** from the first_3 virtual machine **3130** executed by the first host processor **3100**. The processor **1210** may access the register group **1220** in response to the first request REQ**1** received from the first_3 virtual machine **3130**, and may obtain the first_3 virtual function VF**1_3** stored in the register group **1220**.
[0090] The processor **1220** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the first request REQ**1** based on the first_3 virtual function VF**1_3** in response to the first request REQ**1** received from the first_3 virtual machine **3130**. The processor **1210** may control the nonvolatile memory device **1100** to store the data in the first_3 storage area **1113** based on the first_3 virtual function VF**1_3**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the first_3 storage area **1113** based on the first_3 virtual function VF**1_3**.
[0091] In an embodiment, the storage controller **1200** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the first request REQ**1** received from the first host device **3000** by using the first function **1230** including the first physical function PF**1**.
[0092] FIG. **5** shows a storage device for performing an operation that corresponds to a second request of a second host device according to an embodiment.
[0093] Referring to FIG. **5**, the center controller **500** may include a second host device **4000**, a storage controller **1200**, and a nonvolatile memory device **1100**. The second host device **4000** may include a second host processor **4100**. The nonvolatile memory device **1100** may include a second storage area **1120**. The second storage area **1120** may include a second_1 storage area **1121**, a second_2 storage area **1122**, and a second_3 storage area **1123**.
[0094] In an embodiment, the second host device **4000** may have the safety integrity level that is different from that of the first host device **3000**. In an embodiment, the second host device **4000** may have the higher second safety integrity level than the first host device **3000** with the first safety integrity level. For example, the first host device **3000** may correspond to the QM level, and the second host device **4000** may correspond to the ASIL-D.
[0095] In an embodiment, the second host processor **4100** may execute the virtual machines, the operating systems, or the applications. In an embodiment, the second host processor **4100** may execute the second_1 virtual machine **4110**, the second_2 virtual machine **4120**, and the second_3 virtual machine **4130**.
[0096] In an embodiment, the second host processor **4100** may control an operation that corresponds to the second safety integrity level. In an embodiment, the second host processor **4100** may transmit the second request REQ**2** instructing to store or read the data used to the operation that corresponds to the second safety integrity level to the storage controller **1200**. For example, the operation that corresponds to the second safety integrity level may process the data relating to control of the air bag, the anti-lock brake, and the power steering.
[0097] The storage controller **1200** may receive the second request REQ**2** from the second host processor **4100**. In an embodiment, the second request REQ**2** may correspond to the second safety integrity level. For example, the second request REQ**2** that corresponds to the second safety integrity level may instruct to store or read the data relating to control of the air bag, the anti-lock brake, and the power steering.
[0098] The storage controller **1200** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the second request REQ**2** based on the second function **1240** in response to the second request REQ**2** of the second host processor **4100**. In an embodiment, the operation that corresponds to the second request REQ**2** may include a write operation for storing the data received from the second host processor **4100** in the nonvolatile memory device **1100**. The operation that corresponds to the second request REQ**2** may include a read operation for providing

the data read from the nonvolatile memory device **1100** to the second host processor **4100**.

[0099] In an embodiment, the processor **1210** may access the register group **1220** in response to the second request REQ**2**, and may obtain the second function **1240** stored in the register group **1220**. The second function **1240** may include a second physical function PF**2**, a second_1 virtual function VF**2_1**, a second_2 virtual function VF**2_2**, a second_3 virtual function VF**2_3**, and an error detection function **1241**.

[0100] In an embodiment, the processor **1210** may control the nonvolatile memory device **1100** to store the data received from the second host processor **4100** in the second storage area **1120** based on the second physical function PF**2** in response to the second request REQ**2**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the second storage area **1120** based on the second physical function PF**2** in response to the second request REQ**2**.

[0101] In an embodiment, the processor **1210** may receive the second request REQ**2** from the second_1 virtual machine **4110** executed by the second host processor **4100**. The processor **1210** may access the register group **1220** in response to the second request REQ**2** received from the second_1 virtual machine **4110**, and may obtain the second_1 virtual function VF**2_1** stored in the register group **1220**.

[0102] The processor **1210** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the second request REQ**2** based on the second_1 virtual function VF**2_1** in response to the second request REQ**2** received from the second_1 virtual machine **4110**. The processor **1210** may control the nonvolatile memory device **1100** to store the data in the second_1 storage area **1121** that corresponds to the second_1 virtual function VF**2_1**. The data generated by the second_1 virtual machine **4110** may be stored in the second_1 storage area **1121** according to the second request REQ**2**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the second_1 storage area **1121** based on the second_1 virtual function VF**2_1**.

[0103] In an embodiment, the processor **1210** may receive the second request REQ**2** from the second_2 virtual machine **4120** executed by the second host processor **4100**. The processor **1210** may access the register group **1220** in response to the second request REQ**2** received from the second_2 virtual machine **4120**, and may obtain the second_2 virtual function VF**2_2** stored in the register group **1220**.

[0104] The processor **1210** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the second request REQ**2** based on the second_2 virtual function VF**2_2** in response to the second request REQ**2** received from the second_2 virtual machine **4120**. The processor **1210** may control the nonvolatile memory device **1100** to store the data in the second_2 storage area **1122** that corresponds to the second_2 virtual function VF**2_2**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the second_2 storage area **1122** based on the second_2 virtual function VF**2_2**.

[0105] In an embodiment, the processor **1210** may receive the second request REQ**2** from the second_3 virtual machine **4130** executed by the second host processor **4100**. The processor **1210** may access the register group **1220** in response to the second request REQ**2** received from the second_3 virtual machine **4130**, and may obtain the second_3 virtual function VF**2_3** stored in the register group **1220**.

[0106] The processor **1210** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the second request REQ**2** based on the second_3 virtual function VF**2_3** in response to the second request REQ**2** received from the second_3 virtual machine **4130**. The processor **1210** may control the nonvolatile memory device **1100** to store the data in the second_3 storage area **1123** that corresponds to the second_3 virtual function VF**2_3**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the second_3 storage area **1123** based on the second_3 virtual function VF**2_3**.

[0107] In an embodiment, the processor **1210** may obtain the error detection function **1241** from the second register group **1222** in which the second function **1240** is stored from among the register groups included in the register group **1220** in response to the second request REQ**2** received from the second host processor **4100**.

[0108] In an embodiment, the error detection function **1241** may detect the error generated by the second function **1240**. In an embodiment, the error detection function **1241** may detect whether the errors are generated to the second physical function PF**2**, second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3** while the processor **1210** accesses the registers in which the second physical function PF**2**, the second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3** are stored in response to the second request REQ**2**.

[0109] In an embodiment, the error detection function **1241** may perform a recovery operation for correcting the errors generated by the second physical function PF**2**, the second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3**. In an embodiment, the recovery operation may reset the registers in which the second physical function PF**2**, the second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3** are stored, or may reset the storage controller **1200**.

[0110] In an embodiment, the error detection function **1241** may provide information on the errors generated by the second physical function PF**2**, the second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3** to the second host processor **4100**.

[0111] In an embodiment, the error detection function **1241** may control the access to the second register group **1222** storing the second physical function PF**2**, the second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3** based on access right information included in the second request REQ**2**. The access right information may include access allow information indicating that there is a right to access the second register group **1222** or access prohibit information indicating that there is no right to access the second register group **1222**.

[0112] In an embodiment, the error detection function **1241** may allow the access to the second register group **1222** when the access right information included in the second request REQ**2** includes access allow information. The error detection function **1241** may prohibit the access to the second register group **1222** when the access right information included in the second request REQ**2** includes access prohibit information.

[0113] In an embodiment, the storage controller **1200** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the second request REQ**2** received from the second host device **4000** by using the second function **1240** including the second physical function PF**2**.

[0114] In an embodiment, the storage device **1000** may perform the operations that correspond to the first request and the second request received from the first host processor **3100** and the second host processor **4100** with different safety integrity levels by using the first function **1230** and the second function **1240** including different physical functions. The storage device **1000** may perform an operation that corresponds to the second request REQ**2** of the second host processor **4100** by using the second physical function PF**2** when errors are generated to the first physical function PF**1** used to the operation that corresponds to the first request REQ**1** received from the first host processor **3100**.

[0115] In an embodiment, the operation that corresponds to the second request REQ**2** received from the second host processor **4100** may need higher safety than the operation that corresponds to the first request REQ**1** received from the first host processor **3100**. The storage device **1000** may reduce the generation of errors to the operation that corresponds to the second request REQ**2** by detecting and correcting the errors of the second physical function PF**2**, second_1 virtual function VF**2_1**, second_2 virtual function VF**2_2**, and second_3 virtual function VF**2_3** by using the error detection function **1241** included in the second function **1240**.

[0116] FIG. **6** shows a storage device for performing an operation that corresponds to a third request of a third host device according to an embodiment.

[0117] Referring to FIG. **6**, the center controller **500** may include a third host device **5000**, a storage controller **1200**, and a nonvolatile memory device **1100**. The third host device **5000** may include a third host processor **5100**. The nonvolatile memory device **1100** may include a third storage area **1130**. The third storage area **1130** may include a third_1 storage area **1131**, a third_2 storage area **1132**, and a third_3 storage area **1133**.

[0118] In an embodiment, the third host device **5000** may have the safety integrity level that is different from those of the first host device **3000** and the second host device **4000**. In an embodiment, the third host device **5000** may have the third safety integrity level that is higher than the first safety integrity level of the first host device **3000** and is lower than the second safety integrity level of the second host device **4000**. For example, the first host device **3000** may correspond to the QM level, the second host device **4000** may correspond to the ASIL-D, and the third host device **5000** may correspond to the ASIL-B.

[0119] In another embodiment, the third host device **5000** may have the safety integrity level that is higher than that of the first host device **3000**, and may have the same safety integrity level as the second host device **4000**. For example, the first host device **3000** may correspond to the QM level, and the second host device **4000** and the third host device **5000** may correspond to the ASIL-D.

[0120] In an embodiment, the third host processor **5100** may perform the virtual machines, the operating systems, or the applications. In an embodiment, the third host processor **5100** may perform the third_1 virtual machine **5110**, the third_2 virtual machine **5120**, and the third_3 virtual machine **5130**.

[0121] In an embodiment, the third host processor **5100** may control an operation that corresponds to the third safety integrity level. In an embodiment, the third host processor **5100** may transmit the third request REQ**3** for instructing to store or read the data used to the operation that corresponds to the third safety integrity level to the storage controller **1200**. For example, the operation that corresponds to the third safety integrity level may process data relating to a rear-view camera or control such as a stop.

[0122] The storage controller **1200** may receive the third request REQ**3** from the third host processor **5100**. In an embodiment, the third request REQ**3** may correspond to the third safety integrity level. For example, the third request REQ**3** that corresponds to the third safety integrity level may instruct to store or read the data relating to the rear-view camera or control such as a stop. The storage controller **1200** may control the nonvolatile memory device **1100** to perform an operation that corresponds to the third request REQ**3** based on the third function **1250** in response to the third request REQ**3** of the third host processor **5100**.

[0123] In an embodiment, the processor **1210** may access the register group **1220** in response to the third request REQ**3**, and may obtain the third function **1250** stored in the register group **1220**. The third function **1250** may include a third physical function PF**3**, a third_1 virtual function VF**3_1**, a third_2 virtual function VF**3_2**, a third_3 virtual function VF**3_3**, and an error detection function **1251**. In an embodiment, the error detection function **1251** included in the third function **1250** may be realized to be identical to the error detection function **1241** included in the second function **1240**.

[0124] In an embodiment, the processor **1210** may control the nonvolatile memory device **1100** to store the data received from the third host processor **5100** in the third storage area **1130** based on the third physical function PF**3** in response to the third request REQ**3**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the third storage area **1130** based on the third physical function PF**3** in response to the third request REQ**3**.

[0125] In an embodiment, the processor **1210** may access the register group **1220** in response to the third request REQ**3** received from the third_1 virtual machine **5110** executed by the third host processor **5100**, and may obtain the third_1 virtual function VF**3_1** stored in the register group **1220**. The processor **1210** may control the nonvolatile memory device **1100** to store the data in the third_1 storage area **1131** based on the third_1 virtual function VF**3_1** in response to the third request REQ**3** received from the third_1 virtual machine **5110**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the third_1 storage area **1131** based on the third_1 virtual function VF**3_1** in response to the third request REQ**3** received from the third_1 virtual machine **5110**.

[0126] In an embodiment, the processor **1210** may access the register group **1220** in response to the third request REQ**3** received from the third_2 virtual machine **5120** executed by the third host processor **5100**, and may obtain the third_2 virtual function VF**3_2** stored in the register group **1220**.

[0127] The processor **1210** may control the nonvolatile memory device **1100** to store the data in the third_2 storage area **1132** based on the third_2virtual function VF**3_3** in response to the third request REQ**3** received from the third_2 virtual machine **5120**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the third_2 storage area **1132** based on the third_2 virtual function VF**3_2** in response to the third request REQ**3** received from the third_2 virtual machine **5120**.

[0128] In an embodiment, the processor **1210** may access the register group **1220** in response to the third request REQ**3** received from the third_3 virtual machine **5130** executed by the third host processor **5100**, and may obtain the third_3 virtual function VF**3_3** stored in the register group **1220**.

[0129] The processor **1210** may control the nonvolatile memory device **1100** to store the data in the third_3 storage area **1133** based on the third_3 virtual function VF**3_3** in response to the third request REQ**3** received from the third_3 virtual machine **5130**. The processor **1210** may control the nonvolatile memory device **1100** to read the data stored in the third_3 storage area **1133** based on the third_3 virtual function VF**3_3** in response to the third request REQ**3** received from the third_3 virtual machine **5130**.

[0130] In an embodiment, the processor **1210** may obtain the error detection function **1251** from the third register group **1223** storing the third function **1250** from among the register groups included in the register group **1220** in response to the third request REQ**3** received from the second host processor **5100**.

[0131] In an embodiment, the error detection function **1251** may detect the generation of errors of the third function **1250** while the processor accesses the third register group **1223** storing the third function **1250**. The error detection function **1251** may perform the recovery operation for correcting the errors generated by the third function **1250**. The error detection function **1251** may provide information on the errors of the third function **1250** to the third host processor **5100**.

[0132] In an embodiment, the error detection function **1251** may allow or prohibit the access to the third register group **1223** storing the third function **1250** based on the access right information included in the third request REQ**3**.

[0133] In an embodiment, the storage device **1000** may receive the third request REQ**3** from the third host device **5000** with the safety integrity level that is different from those of the first host device **3000** and the second host device **4000**, and may perform an operation that corresponds to the third request REQ**3** by using the third function **1250** including the third physical function PF**3** that is different from the first physical function PF**1** and the second physical function PF**2** used to the first request REQ**1** and the second request REQ**2** of the first host device **3000** and the second host device **4000**.

[0134] In an embodiment, the storage device **1000** may increase the safety of the operation that corresponds to the third request REQ**3** by detecting and correcting the error of the third function **1250** generated while accessing the third register group **1223** storing the third function **1250** by use of the error detection function **1251** included in the third function **1250**.

[0135] FIG. **7** shows a register including a bit flip correcting circuit according to an embodiment.
[0136] Referring to FIG. **7**, the storage controller **1200** may include a processor **1210** and a second_1 register REGISTER **2_1**. The second_1 register REGISTER **2_1** may be included in the register group **1220** of FIG. **3**.
[0137] The second_1 register REGISTER **2_1** may include flipflops **1224** and a bit flip correcting circuit **1225**. The flipflops **1224** may store the second physical function PF**2**. The bit flip correcting circuit **1225** may detect and correct bit flips of the second physical function PF**2**.
[0138] In an embodiment, the processor **1210** may transmit the command CMD for obtaining the second physical function PF**2** stored in the flipflops **1224** to the second_1 register REGISTER **2_1**. The second_1 register REGISTER **2_1** may provide the second physical function PF**2** stored in the flipflops **1224** to the processor **1210** in response to the command CMD. The processor **1210** may control the operation that corresponds to the second request REQ**2** of the second host processor **4100** by using the second physical function PF**2**.
[0139] In an embodiment, the second physical function PF**2** stored in the flipflops **1224** may generate bit flips by which bit values indicating the second physical function PF**2** are changed to 0 from 1 or to 1 from 0 according to noise generated while being read in response to the command CMD.
[0140] In an embodiment, the bit flip correcting circuit **1225** may detect the bit flip generated while the second physical function PF**2** stored in the flipflops **1224** is read, and may correct the bit flip. The processor **1210** may receive the second physical function PF**2** of which the bit flip is corrected by the bit flip correcting circuit **1225**.
[0141] FIG. **7** has described the second_1 register REGISTER **2_1**, and the respective registers included in the second register group **1222** and the third register group **1223** may include the bit flip correcting circuit **1225** in a like way of the second_1 register REGISTER **2_1**.
[0142] FIG. **8** shows a register group for storing a mirror function according to an embodiment.
[0143] The first register group **1221** will not be described in FIG. **8**. Referring to FIG. **8**, the storage controller **1200** may include a register group **1220**. The register group **1220** may include a second register group **1222** and a third register group **1223**. The second register group **1222** and the third register group **1223** may include registers.
[0144] In an embodiment, the second register group **1222** may store the second function **1240**. The second function **1240** may include a second physical function PF**2**, a second_1 virtual function VF**2_1**, a second_2 virtual function VF**2_2**, a second_3 virtual function VF**2_3**, an error detection function **1241**, and a first mirror function **1242**. The first mirror function **1242** may be a function generated by mirroring the third physical function PF**3** included in the third function **1250**, the third_1 virtual function VF**3_1**, the third_2 virtual function VF**3_2**, and the third_3 virtual function VF**3_3**.
[0145] In an embodiment, the first mirror function **1242** may include a third physical mirror function PF**3**′ generated by mirroring the third physical function PF**3**, a third_1 virtual mirror function VF**3_1**′ generated by mirroring the third_1 virtual function VF**3_1**, a third_2 virtual mirror function VF**3_2**′ generated by mirroring the third_2 virtual function VF**3_2**, and a third_3 virtual mirror function VF**3_3**′ generated by mirroring the third_3 virtual function VF**3_3**.
[0146] In an embodiment, the third register group **1223** may store the third function **1250**. The third function **1250** may include a third physical function PF**3**, a third_1 virtual function VF**3_1**, a third_2 virtual function VF**3_2**, a third_3 virtual function VF**3_3**, an error detection function **1251**, and a second mirror function **1252**. The second mirror function **1252** may be a function generated by mirroring the second physical function PF**2** included in the second function **1240**, the second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3**.
[0147] In an embodiment, the second mirror function **1252** may include a second physical mirror function PF**2**′ generated by mirroring the second physical function PF**2**, a second_1 virtual mirror

function VF**2_1**′ generated by mirroring the second_1 virtual function VF**2_1**, a second_2 virtual mirror function VF**2_2**′ generated by mirroring the second_2 virtual function VF**2_2**, and a second_3 virtual mirror function VF**2_3**′ generated by mirroring the second_3 virtual function VF**2_3**.

[0148] In an embodiment, the storage controller **1200** may detect generation of errors of the third function **1250** by using the first mirror function **1242** included in the second function **1240**. In an embodiment, the processor **1210** may detect the generation of errors of the third function **1250** based on a result of comparing the third physical function PF**3**, the third_1 virtual function VF**3_1**, the third_2 virtual function VF**3_2**, and the third_3 virtual function VF**3_3** included in the third function **1250** and the third physical mirror function PF**3**′, the third_1 virtual mirror function VF**3_1**′, the third_2 virtual mirror function VF**3_2**′, and the third_3 virtual mirror function VF**3_3**′ included in the first mirror function **1242**.

[0149] In an embodiment, the storage controller **1200** may detect the generation of errors of the second function **1240** by using the second mirror function **1252** included in the third function **1250**. In an embodiment, the processor **1210** may detect the generation of errors of the second function **1240** based on a result of comparing the second physical function PF**2**, the second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3** included in the second function **1240** and the second physical mirror function PF**2**′, the second_1 virtual mirror function VF**2_1**′, the second_2 virtual mirror function VF**2_2**′, and the second_3 virtual mirror function VF**2_3**′ included in the second mirror function **1252**.

[0150] FIG. **9** shows a storage device for detecting errors of a function by use of a mirror function according to an embodiment.

[0151] Referring to FIG. **9**, the center controller **500** may include a second host device **4000**, a storage controller **1200**, and a nonvolatile memory device **1100**.

[0152] In an embodiment, the processor **1210** may control the nonvolatile memory device **1100** to obtain the second function **1240** stored in the register group **1220** in response to the second request received from the second host processor **4100**, and perform a write operation or a read operation on the second storage area **1120** by using the second physical function PF**2**, the second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3** included in the second function **1240**.

[0153] In an embodiment, the processor **1210** may access the register group **1220**, and may obtain the second mirror function included in the third function **1250**. The second mirror function may include a second physical mirror function PF**2**′ generated by mirroring the second physical function PF**2**, a second_1 virtual mirror function VF**2_1**′ generated by mirroring the second_1 virtual function VF**2_1**, a second_2 virtual mirror function VF**2_2**′ generated by mirroring the second_2 virtual function VF**2_2**, and a second_3 virtual mirror function VF**2_3**′ generated by mirroring the second_3 virtual function VF**2_3**.

[0154] In an embodiment, the error detect module **1241** may detect whether errors are generated to the second physical function PF**2**, the second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3** while the processor **1210** accesses the register group **1220**. In an embodiment, the error detect module **1241** may detect the generation of errors of the second physical function PF**2**, the second_1 virtual function VF**2_1**, the second_2 virtual function VF**2_2**, and the second_3 virtual function VF**2_3** by using the second mirror function included in the third function **1250**.

[0155] In an embodiment, the error detect module **1241** may detect the error of the second physical function PF**2** based on a result of comparing the second physical function PF**2** and the second physical mirror function PF**2**′. The error detect module **1241** may detect the error of the second_1 virtual function VF**2_1** based on a result of comparing the second_1 virtual function VF**2_1** and the second_1 virtual mirror function VF**2_1**′. The error detect module **1241** may detect the error of the second_2 virtual function VF**2_2** based on a result of comparing the second_2 virtual function

VF**2_2** and the second_2 virtual mirror function VF**2_2**′. The error detect module **1241** may detect the error of the second_3 virtual function VF**2_3** based on a result of comparing the second_3 virtual function VF**2_3** and the second_3 virtual mirror function VF**2_3**′.

[0156] FIG. **10** shows a flowchart of a storage device for performing an operation that corresponds to a request of a host device by using functions including physical functions according to an embodiment.

[0157] Referring to FIG. **10**, at S**10**, the storage device **1000** may receive a first request from the first host device **3000**. In an embodiment, the first host device **3000** may have the first safety integrity level. For example, the first host device **3000** may correspond to the QM level. For example, the first host device **3000** may perform the operation for processing the data relating to the in-vehicle infotainment.

[0158] At S**13**, the storage device **1000** may perform a first operation that corresponds to the first request by using the first function including the first physical function. The first operation may store the data received from the first host device **3000** or may provide the data to the first host device **3000**.

[0159] At S**15**, the storage device **1000** may receive a second request from the second host device **4000**. In an embodiment, the second host device **4000** may have the second safety integrity level that is higher than the first safety integrity level. The second host device **4000** may perform an operation that needs higher safety than the operation performed by the first host device **3000**. For example, the second host device **4000** may correspond to the ASIL-D. For example, the second host device **4000** may perform an operation for processing the data relating to the ADAS.

[0160] At S**17**, the storage device **1000** may perform a second operation that correspond to the second request by using the second function including the second physical function. The second operation may store the data received from the second host device **4000** or may provide the data to the second host device **4000**.

[0161] At S**19**, the storage device **1000** may detect the error of the second function. The storage device **1000** may provide information on the error of the second function to the second host device **4000**.

[0162] FIG. **11** shows a configuration of a storage controller according to an embodiment.

[0163] Referring to FIG. **11**, the storage controller **6000** may include a processor **6010**, a RAM **6020**, an error correction circuit **6030**, a host interface **6040**, a register **6050**, and a memory interface **6060**.

[0164] The processor **6010** may control a general operation of the storage controller **6000**. The processor **6010** may control the operation of the storage controller **6000** to store the data requested by the first host device **3000**, the second host device **4000**, and the third host device **5000** in the nonvolatile memory device **1100**.

[0165] The RAM **6020** may be used as a buffer memory, a cache memory, or an operation memory of the storage controller **6000**. In an embodiment, the RAM **6020** may temporarily store the data received from the first host device **3000**, the second host device **4000**, and the third host device **5000**.

[0166] The error correction circuit **6030** may perform an error correction operation. In an embodiment, the error correction circuit **6030** may perform an error correction code (ECC) encoding on the data to be stored in the nonvolatile memory device **1100** through the memory interface **6060**. The error correction encoded data may be transmitted to the nonvolatile memory device **1100** through the memory interface **6060**. In an embodiment, the error correction circuit **6030** may perform a ECC decoding on the data received from the nonvolatile memory device **1100**.

[0167] The register **6050** may store the first function including the first physical function, the second function including the second physical function, and the third function including the third physical function. The first function may be used in performing the operation that corresponds to the first request received from the first host device **3000**. The second function may be used in performing an operation that corresponds to the second request received from the second host device **4000**. The third function may be used in performing the operation that corresponds to the third request received from the third host device **5000**.

[0168] The storage controller **6000** may communicate with the PCIe switch **2000** through the host interface **6040**. The host interface **6040** may transmit/receive data to/from the first host device **3000**, the second host device **4000**, and the third host device **5000** through the PCIe switch **2000**.

[0169] The storage controller **6000** may communicate with the nonvolatile memory device **1100** through the memory interface **6060**. The storage controller **6000** may provide commands, addresses, and data to the nonvolatile memory device **1100** through the memory interface **6060**.

[0170] FIG. **12** shows a configuration of a nonvolatile memory device according to an embodiment.

[0171] Referring to FIG. **12**, the nonvolatile memory device **1100** may include a memory cell array **110**, a voltage generator **120**, a row decoder **130**, a page buffer group **140**, and a control circuit **150**.

[0172] The memory cell array **110** may include memory blocks BLK**1** to BLKz. The memory blocks BLK**1** to BLKz may be connected to the row decoder **130** through the row lines RL. The memory blocks BLK**1** to BLKz may be connected to the page buffer group **140** through the bit lines BL. In an embodiment, the memory blocks BLK**1** to BLKz may include a first storage area, a second storage area, and a third storage area.

[0173] The respective memory blocks BLK**1** to BLKz may include memory cells. In an embodiment, the memory cells may be nonvolatile memory cells.

[0174] The voltage generator **120** may generate operating voltages Vop by using an external power voltage supplied to the nonvolatile memory device **1100**. The voltage generator **120** may be operated in response to control by the control circuit **150**.

[0175] In an embodiment, the voltage generator **120** may generate the operating voltages Vop used to the program operation, the read operation, and the erase operation. For example, the voltage generator **120** may generate an erase voltage, a program voltage, a pass voltage, and a read voltage. The operating voltages Vop may be supplied to the memory cell array **110** by the row decoder **130**.

[0176] The row decoder **130** may be connected to the memory cell array **110** through the row lines RL. The row lines RL may include drain selecting lines, word lines, and source selecting lines.

[0177] The row decoder **130** may be operated in response to the control by the control circuit **150**. The row decoder **130** may receive a row address X-ADDR from the control circuit **150**. In an embodiment, the row decoder **130** may select at least one of the word lines based on the row address X-ADDR, and may apply the operating voltages Vop provided by the voltage generator **120** to the at least one word line.

[0178] In an embodiment, the row decoder **130** may apply the program voltage to the selected one of the word lines and may apply the pass voltage that has a lower level than the program voltage to the non-word lines during the program operation. The row decoder **130** may apply a verify voltage to the selected word line and may apply a verification pass voltage that has the higher level than a verifying voltage to the non-selected word lines during the program verifying operation.

[0179] The row decoder **130** may apply the read voltage to the selected word line and may apply a read pass voltage that has the higher level than the read voltage to the non-selected word lines during the read operation.

[0180] The page buffer group **140** may include page buffers PB**1** to PBn. The page buffers PB**1** to PBn may be respective connected to the memory cell array **110** through the bit lines BL. The page buffers PB**1** to PBn may be operated in response to the control by the control circuit **150**.

[0181] In an embodiment, the page buffers PB**1** to PBn may receive data DATA from outside the memory device. The page buffers PB**1** to PBn may select at least one of the bit lines BL based on a column address Y-ADDR received from the control circuit **150**.

[0182] In an embodiment, the page buffers PB**1** to PBn may transmit the data received from outside the memory device to the memory cells of the memory cell array **110** through the bit lines

BL during the program operation. The memory cells may be programmed according to the received data. The page buffers PB**1** to PBn may sense the data stored in the memory cells through the bit lines BL during the program verifying operation.
[0183] The page buffers PB**1** to PBn may sense the data stored in the memory cells through the bit lines BL and may store the sensed data in the page buffers PB**1** to PBn during the read operation.
[0184] The control circuit **150** may be connected to the voltage generator **120**, the row decoder **130**, and the page buffer group **140**. The control circuit **150** may control the general operation of the nonvolatile memory device **1100**. The control circuit **150** may be operated in response to the command CMD transmitted from the outside. The control circuit **150** may generate various signals in response to the command CMD and the address ADDR may control the voltage generator **120**, the row decoder **130**, and the page buffer group **140**.
[0185] Although an embodiment of the present disclosure has been described in detail above, the scope of the present disclosure is not limited thereto, and a person of an ordinary skill in using the basic concept of the present disclosure defined in the following claims range Various modifications and improvements of the art also belong to the scope of the present disclosure. What is claimed is:

Claims

1. An automotive storage device comprising: a nonvolatile memory device configured to store data; and a storage controller configured to receive a first request from a first host, control the nonvolatile memory device so that the nonvolatile memory device performs an operation corresponding to the first request by using a first function including a first physical function, receive a second request from a second host having a safety integrity level that is different from that of the first host, and control the nonvolatile memory device so that the nonvolatile memory device performs an operation corresponding to the second request by using a second function including a second physical function.
2. The automotive storage device of claim 1, wherein the second function includes an error detection function for detecting generation of errors of the second function.
3. The automotive storage device of claim 2, wherein the error detection function is configured to allow access to registers storing the second function based on access right information included in the second request.
4. The automotive storage device of claim 1, wherein the storage controller includes registers storing the second function, and the registers include bit flip correcting circuits for detecting bit flips of the second function.
5. The automotive storage device of claim 1, wherein an operation corresponding to the first request stores or reads data relating to an in-vehicle infotainment, and an operation corresponding to the second request corresponds to a safety integrity level that is higher than that of the first host, the operation corresponding to the first request, and stores or reads data relating to control of an air bag, an anti-lock brake, and a power steering of a vehicle.
6. The automotive storage device of claim 1, wherein the storage controller is configured to receive a third request from a third host having a safety integrity level that is different from that of the first host, and control the nonvolatile memory device so that the nonvolatile memory device performs an operation corresponding to the third request by using a third function including a third physical function.
7. The automotive storage device of claim 6, wherein the operation corresponding to the third request corresponds to a safety integrity level that is lower than that of an operation corresponding to the second request, and stores or reads data relating to a rear-view camera of a vehicle or control of a stop.
8. The automotive storage device of claim 6, wherein the third function includes a mirror function generated by mirroring the second function, and the storage controller is configured to detect an

error of the second function based on a result of comparing the second function and the mirror function.
9. The automotive storage device of claim 1, wherein the first function includes a first virtual function, the second function includes a second virtual function, the first virtual function corresponds to a first virtual machine executed by the first host, and the second virtual function corresponds to a second virtual machine executed by the second host.
10. The automotive storage device of claim 9, wherein the nonvolatile memory device includes storage areas, and the storage controller is configured to control the nonvolatile memory device to store the data generated by the second virtual machine in a storage area corresponding to the second virtual function from among the storage areas based on the second virtual function in response to the second request.
11. An automotive storage controller comprising: first registers configured to store a first function including a first physical function; second registers configured to store a second function including a second physical function; and a processor configured to receive a first request corresponding to a first safety integrity level from outside the automotive storage controller, generate a command for performing an operation corresponding to the first request based on the first function, receive a second request corresponding to a second safety integrity level from outside the automotive storage controller, generate a command for performing an operation corresponding to the second request based on the second function, and detect generation of errors of the second function while accessing the second registers.
12. The automotive storage controller of claim 11, wherein the second function includes an error detection function for allowing an access to the second registers based on access right information included in the second request.
13. The automotive storage controller of claim 11, further comprising third registers configured to store a third function including a third physical function, wherein the processor is configured to receive a third request corresponding to a third safety integrity level that is lower than the second safety integrity level from outside the automotive storage controller, and generate a command for performing an operation corresponding to the third request based on the third function.
14. The automotive storage controller of claim 13, wherein the operation corresponding to the second request stores or reads the data relating to control of an air bag, an anti-lock brake, and a power steering of a vehicle, and the operation corresponding to the third request stores or read the data relating to a rear-view camera of the vehicle or control of a stop.
15. The automotive storage controller of claim 13, wherein the second function includes a mirror function generated by mirroring the third function, and the processor is configured to detect an error of the third function based on a result of comparing the third function and the mirror function.
16. An automotive electronic system comprising: a first host processor with a first safety integrity level; a second host processor with a second safety integrity level that is higher than the first safety integrity level; and an automotive storage device configured to perform an operation corresponding to a first request of the first host processor based on a first function including a first physical function in response to the first request, and perform an operation corresponding to a second request of the second host processor based on a second function including a second physical function in response to the second request.
17. The automotive electronic system of claim 16, wherein the automotive storage device includes a storage controller configured to detect an error of the second function while performing an operation corresponding to the second request.
18. The automotive electronic system of claim 16, further comprising a third host processor with the second safety integrity level, wherein the automotive storage device is configured to receive a third request of the third host processor, and access registers storing a third function including a third physical function based on access right information included in the third request.
19. The automotive electronic system of claim 16, wherein the first function includes first virtual

functions, the second function includes second virtual functions, the first virtual functions correspond to operating systems or applications performed by the first host processor, and the second virtual functions correspond to operating systems or applications performed by the second host processor.

20. The automotive electronic system of claim 16, wherein the first host processor is configured to process data relating to an automotive infotainment, and the second host processor is configured to process data relating to an advanced driver assistance system (ADAS).